IIITB COMET FOUNDATION
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GATE CS 2019 – Question 6 Analysis

Question:

In 16-bit 2's complement representation, the decimal number -28 is:

(A) 1111 1111 0001 1100 (B) 0000 0000 1110 0100 (C) 1111 1111 1110 0100 (D) 1000 0000 1110 0100

Step 1: Convert +28 to Binary

$$28_{10} = 11100_2$$

16-bit form:

$$0000\ 0000\ 0001\ 1100$$

Step 2: 1's Complement

$$1111\ 1111\ 1110\ 0011$$

Step 3: Add 1

$$1111\ 1111\ 1110\ 0011 + 0000\ 0000\ 0000\ 0001 = \mathbf{1111\ 1111\ 1110\ 0100}$$

Correct Option: (C)

Hardware Implementation

Verified using Arduino UNO.

PlatformIO Code:

```
#include <Arduino.h>
#include <stdint.h>

int16_t num = 28;
int16_t twos_comp;

void setup() {
    twos_comp = -num;
}
```

```
void loop() {
}
```

Components Used:

- Arduino UNO
- Breadboard
- Common Anode 7-Segment Display
- 220Ω Resistor
- Jumper Wires

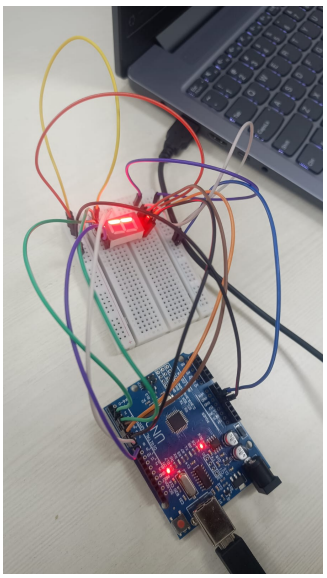
Pin Connections:

Segment a	Digital 2
Segment b	Digital 3
Segment c	Digital 4
Segment d	Digital 5
Segment e	Digital 6
Segment f	Digital 7
Segment g	Digital 8
GND	GND
VCC	5V

Observation:

The display showed **1**, confirming correctness.

Hardware Output



Conclusion:

The 16-bit 2's complement of -28 is:

$$1111\ 1111\ 1110\ 0100$$

Hardware and theoretical results match.